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The anti-inflammatory effects of palmitoylethanolamide (PEA) on endotoxin-induced uveitis in rats

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ABSTRACT

The aim of this study was to investigate the effects of palmitoylethanolamide (PEA), an endogenous fatty acid amide belonging to the family of the N-acylethanolamines (NAEs), in rats subjected to endotoxin-induced uveitis (EIU). EIU was induced in male rats by a single footpad injection of 200 μg lipopolysaccharide (LPS). PEA was administered intraperitoneally at 1 h before and 7 h after injection of LPS. Another group of animals was treated with vehicle. Dexamethasone (DEX) was administered as a positive control. Rats were sacrificed 16 h after injection and the eyes tissues were collected for histology, immunohistochemical and western blot analyses. The histological evaluation of the iris–ciliary body showed an increase of neutrophilic infiltration and nuclear modification of vessel of endothelial cells. PEA treatment decreased the inflammatory cell infiltration and improved histological damage of eye tissues. In addition, PEA treatment reduced pro-inflammatory tumor necrosis factor (TNF-α) levels, protein extravasion and lipid peroxidation. Immunohistochemical analysis for intracellular adhesion molecule (ICAM)-1 and nitrotyrosine showed a positive staining from LPS-injected rats. The degree of staining for ICAM-1 and nitrotyrosine was significantly reduced in eye sections from LPS-injected rats treated with PEA. In addition, an increase of inducible nitric oxide synthase (iNOS) and nuclear factor (NF-κB) was also evaluated in inflammed ocular tissues by western blot. PEA strongly inhibited iNOS expression and nuclear NF-κB translocation. Thus, in this study we demonstrated that PEA reduces the degree of ocular inflammation in a rat model of EIU.

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1. Introduction

Anterior uveitis is an inflammation of the iris and parts of the ciliary body. The main therapies for uveitis are corticosteroids but some patients are still resistant to them (Bucolo et al., 2003). Endotoxin induced uveitis (EIU) is a useful animal model for acute anterior uveitis in humans but the cellular mechanisms activated by lipopolysaccharide (LPS) injection are still not clear (Bucolo et al., 2003). Activation of signaling pathways and alterations of the expression of various inflammatory genes are common characteristics in ocular inflamed tissues (Kalariya et al., 2010; Kim and Moudgil, 2008). Activation of nuclear factor (NF-κB) is known to mediate inflammatory diseases, including uveitis (Kalariya et al., 2010). Because of uveal inflammation, the levels of pro-inflammatory cytokines, chemokines and adhesion molecules in ocular tissues increase significantly (Bucolo et al., 2003). Several studies demonstrated that injection of different cytokines such as tumor necrosis factor (TNF)-α, interleukin (IL)-1, IL-6, IL-8, into rat eyes can cause acute uveitis (Chiou et al., 2000; Lemaitre et al., 2001). In addition, high levels of TNF-α were also found in the aqueous humor after LPS injection (Smith et al., 1998). In that regard, it was shown an increased intercellular adhesion molecule 1 (ICAM-1) expression on the vascular endothelium of the rat ciliary body during EIU (Bucolo et al., 2003). Inflammation has a strong correlation with the oxidative stress. Among free radicals, nitric oxide (NO) has the potential to mediate the vascular dysfunction associated with EIU (Goureau et al., 1995). Previous studies demonstrated that the NO-related injury may be due to peroxynitrite formation (Szabo, 1996).

N-palmitoylethanolamide (PEA) is an endogenous fatty acid amide belonging to the family of the N-acylethanolamines (NAEs). PEA has an important analgesic, anti-inflammatory and neuroprotective action, acting at different molecular targets such as immune cells (Hoareau et al., 2006). Several studies demonstrated that the endogenous tissue

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levels of PEA are altered during inflammation (Re et al., 2007). In human eyes PEA levels decrease significantly in the choroid and ciliary body of patients with glaucoma (Chen et al., 2005).

Several mechanisms could explain the anti-inflammatory and anti-hyperalgesic effects of PEA, in particular the activation of a cell surface receptor cannabinoid CB2-like or the orphan GPR55 receptor, or a nuclear receptor of the peroxisome proliferator-activated receptors (PPARs) family (Farquhar-Smith et al., 2002) and the down-modulation of mast cell (MC) degranulation (Autacoid Local Inflammation Antagonism "ALIA" mechanism) (Aloe et al., 1993). Anti-inflammatory and protective activities of PEA were demonstrated in several models of inflammation, such as carrageenan-induced paw edema, adjuvant-induced arthritis, tuberculin hypersensitivity and ischemia reperfusion injury (Di Paola et al., 2012a, 2012b; Esposito and Cuzzocrea, 2013). In addition, PEA exhibits an anti-inflammatory action by the inhibiting of TNF-α levels in the serum of mice treated with LPS and on human adipocytes (Hoareau et al., 2009).

Thus, the aim of the study was to investigate the effects of PEA on the EIU induced by LPS injection in the rat.

2. Materials and methods

2.1. Animals

Male Lewis rats (160–180 gr) were obtained from Harlan Nossan, Italy. All the animals were treated according to the ARVO Statement for the Use of Animals in Ophthalmic and Vision Research.

2.2. EIU induction, protein concentration and TNF-α levels

Uveitis was induced in rats by foot injection of 200 μg LPS (*Salmonella minnesota*). PEA (10 mg/kg, 10% ethanol) was injected intraperitoneally 1 h before and 7 h after LPS injection. Control group received LPS injection and vehicle (ethanol), 1 h before and 7 h after LPS injection. DEX (1 mg/kg) was also administered as a positive control 1 h before and 7 h after LPS injection (Mangano et al., 2008). Sham animals were not injected by LPS but only treated with vehicle or PEA or DEX. Uveitis in each eye was graded using a scoring system as previously reported (Behar-Cohen et al., 1998): 0=no inflammatory reaction; 1=discrete dilatation of the iris and conjunctival vessels; 2=moderate dilatation of the iris and conjunctival vessels; 3=intense iridal hyperemia with flare in anterior chamber; 4=same clinical signs as 3 plus the presence of fibrinous exudate in the pupillary area with intense flare in the anterior chamber. Sixteen hours after uveitis induction, the rats were killed and aqueous humor and iris–ciliary body of right and left eyes were collected. Aqueous humor samples were pooled and the protein concentration was detected by the Bradford assay. In a separate set of animals, TNF-α levels were measured in aqueous humor at 16 h after LPS injection. The assay was carried out by using a ELISA, commercial kit (Calbiochem-Novabiochem Corporation, USA).

2.3. Histological examination

Sixteen hours after uveitis induction, eye tissues were collected and fixed in a solution 0.1% glutareldehyde (25%) and 4% paraformaldehyde for 24 h at room temperature, dehydrated by graded ethanol and embedded in paraplast (Sherwood Medical, Mahwh, NJ, USA). The ocular sections (7 μm) were deparaffinized with xylene, stained with May-Grunwald–Giemsa and studied using light microscopy (Dialux 22 Leitz).

2.4. Evaluation of infiltrating cells and myeloperoxidase activity

The number of neutrophils was counted on histological sections of the iris–ciliary body by an expert unaware of the treatment groups. For each animal, the number of neutrophils is the mean of three serial sections. Myeloperoxidase (MPO) activity, a marker of neutrophil infiltration, was measured as previously indicated (Bucolo et al., 2003). Briefly, 16 h after LPS, eye tissues were weighed and homogenized in a solution containing 0.5% (wt/vol) hexadecyltrimethylammonium bromide dissolved in 10 mmol/l potassium phosphate buffer (pH 7.4) and centrifuged for 30 min at 20,000g at 4 °C. An aliquot of supernatant was added to a reaction mixture containing 1.6 mmol/l tetramethylbenzidine and 0.1 mmol/l hydrogen peroxide (H_2O_2). The rate of change in absorbance was detected spectrophotometrically at 650 nm. MPO activity was defined as the quantity of enzyme required degrading 1 mmol of H_2O_2 at 37 °C and was expressed in U/g wet tissue.

2.5. Malondialdehyde (MDA) measurement

Levels of MDA into the eyes tissues were determined as an index of lipid peroxidation (Talero et al., 2012). Briefly, eye tissues were weighed and homogenized in a 1.15% (wt/vol) KCl solution. 100 μl aliquots of homogenate were added to a reaction mixture containing 200 μl 8.1% (wt/vol) lauryl sulfate, 1.5 ml 20% (vol/vol) acetic acid (pH 3.5), 1.5 ml 0.8% (wt/vol) thiobarbituric acid, and 700 μl distilled water. Samples were then boiled for 1 h at 95 °C and centrifuged at 3000g for 10 min. The absorbance of the supernatant was measured spectrophotometrically at 532 nm. MDA levels were expressed as mM /100 mg wet tissue.

2.6. Immunohistochemical localization of ICAM-1 and nitrotyrosine

After 16 h from the injection of LPS, iris–ciliary bodies were collected and fixed in 10% (w/v) PBS-buffered formaldehyde and 7 μm sections were prepared from paraffin embedded tissues. After deparaffinization, endogenous peroxidase was quenched with 0.3% (v/v) hydrogen peroxide in 60% (v/v) methanol for 30 min. The sections were permeabilized with 0.1% (w/v) Triton X-100 in PBS for 20 min. Non-specific adsorption was minimized by incubating the sections in 2% (v/v) normal goat serum in PBS for 20 min. Endogenous biotin or avidin binding sites were blocked by sequential incubation for 15 min with biotin and avidin (Vector Laboratories, Burlingame, CA), respectively. Tissue sections were incubated overnight with: (1) purified hamster anti-mouse ICAM-1 (CD54) (AbD Serotec, MCA1371Z, 1:500 in PBS, w/v) or (2) anti-nitrotyrosine antibody (Millipore, 06-284). Sections were washed with PBS and incubated with secondary antibody. Specific labeling was determined with a biotin-conjugated goat anti-rabbit IgG and avidin–biotin peroxidase complex (Vector Laboratories, Burlingame, CA). The counter stain was developed with diaminobenzidine (brown color) and nuclear fast red (red background). Positive staining (brown color) was found in the sections, indicating that the immunoreactions were positive. To verify the binding specificity for ICAM-1 and nitrotyrosine some sections were also incubated with primary antibody only (no secondary antibody) or with secondary antibody only (no primary antibody). In these situations, no positive staining was found in the sections. Immunohistochemistry photographs were assessed by densitometry by using Optilab Graftek software on a Macintosh personal computer.

2.7. Western blot analysis for IκBα, NF-κB p65 and iNOS

After 16 h from the injection of LPS, tissue samples from iris–ciliary bodies were suspended in extraction Buffer A containing 0.2 mM PMSF, 0,15 μM pepstatin A, 20 μM leupeptin, 1 mM

sodium orthovanadate, homogenized for 2 min and centrifuged at 1000*g* for 10 min at 4 °C. Supernatants represented the cytosolic fraction. The pellets, containing enriched nuclei, were re-suspended in Buffer B containing 1% Triton X-100, 150 mM NaCl, 10 mM Tris–HCl pH 7.4, 1 mM EGTA, 1 mM EDTA, 0,2 mM PMSF, 20 μm leupeptin, 0,2 mM sodium orthovanadate. After centrifugation for 30 min at 15,000*g* at 4 °C, the supernatants containing the nuclear proteins were stored at −80 °C for further analysis. The levels of IκB-α and iNOS were quantified in cytosolic fraction, while NF-κB p65 levels were quantified in nuclear fraction. The filters were blocked with 1× PBS, 5% (w/v) non fat dried milk (PM) for 40 min at room temperature and then, probed with specific Abs IκB-α (Santa Cruz Biotechnology, 1:1000), or anti-NF-kB p65 (1:1000; Santa Cruz Biotechnology) or anti-iNOS (1:1000; Transduction) in 1× PBS, 5% w/v non fat dried milk, 0.1% Tween-20 (PMT) at 4 °C, overnight. Membranes were incubated with peroxidase-conjugated bovine anti-mouse IgG secondary antibody or peroxidase-conjugated goat anti-rabbit IgG (1:2000, Jackson ImmunoResearch, West Grove, PA) for 1 h at room temperature. To ascertain that blots were loaded with equal amounts of proteins they were also incubated with antibody against β-actin protein (1:10,000 Sigma-Aldrich Corp.). The relative expression of the protein bands of IκB-α (~37 kDa), NF-κB p65 (65 kDa), iNOS (130 kDa) was quantified by densitometric scanning of the X-ray films with GS-700 Imaging Densitometer (GS-700, Bio-Rad Laboratories, Milan, Italy) and a computer program (Molecular Analyst, IBM), and standardized for densitometric analysis to β-actin levels.

2.8. Materials

PEA was provided from Epitech Group S.r.l. (Saccolongo, Italy). All compounds were from Sigma-Aldrich (Milan, Italy). All chemicals were of the highest commercial grade available. All stock solutions were prepared in non-pyrogenic saline (0.9% NaCl; Baxter, Italy, UK).

2.9. Statistical analysis

All values are expressed as mean ± S.D. The results were analyzed by one-way ANOVA followed by a Bonferroni *post-hoc* test for multiple comparisons. A Mann–Withney U test was used for ordinal variables. A value of $P < 0.05$ was considered significant. The figures from histological and immunohistochemical

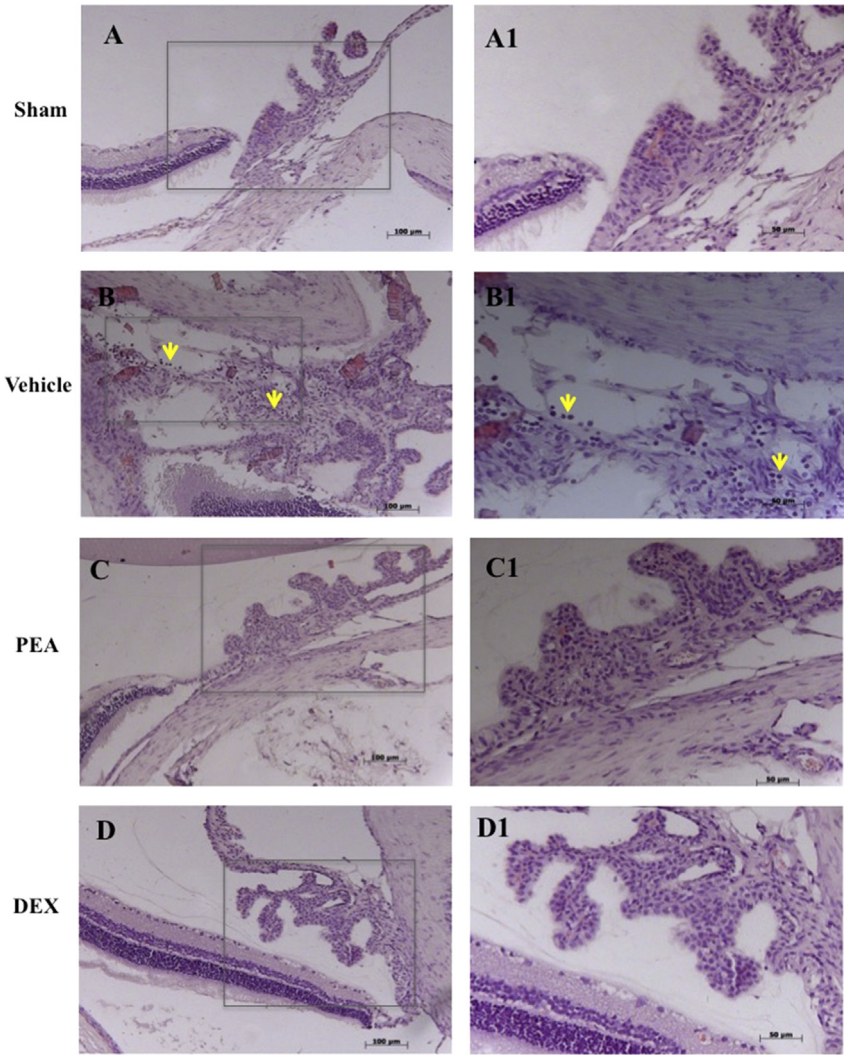


Fig. 1. The effects of PEA on histological appearance of uveitis. Histological appearance of the iris ciliary body 16 h after LPS injection. Vehicle (B, B1), PEA (C, C1), dexamethasone (DEX, positive control) (D, D1) and Sham (A, A1). Yellow arrows mark the infiltrate in LPS + vehicle group (B, B1). This figure is representative of at least 3 experiments performed on different experimental days. For the magnification scale 10×, scale bar is 100 μm (A–C); for the magnification scale 20×, scale bar is 50 μm (A1, B1,C1). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

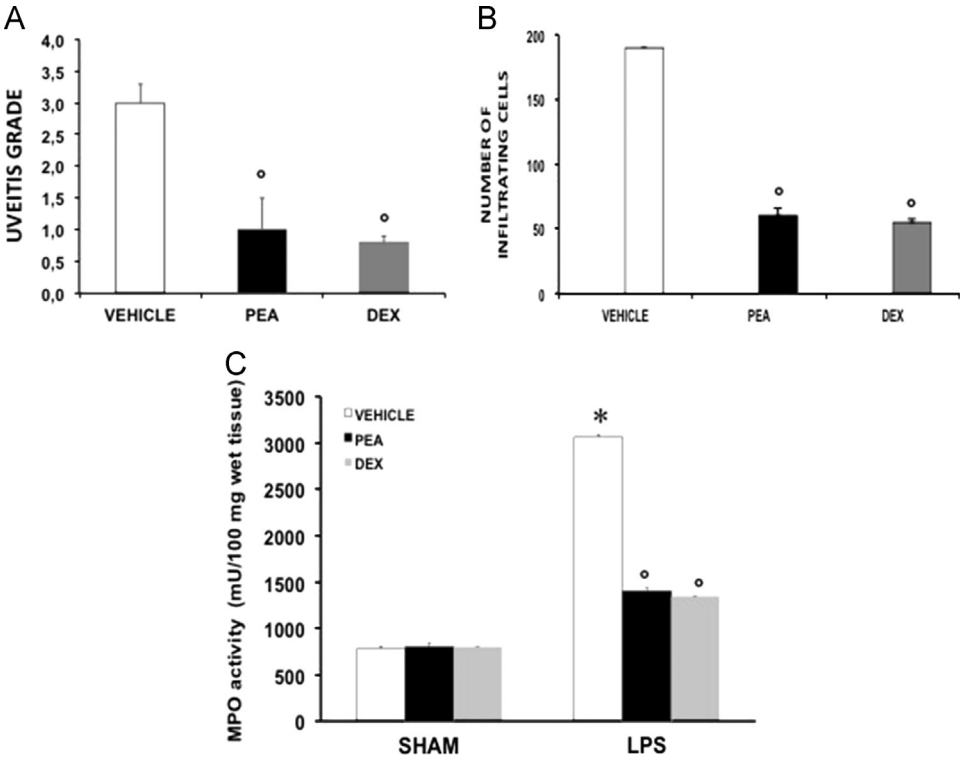


Fig. 2. Effect of PEA on EIU and clinical score. The degree of uveitis and the number of inflammatory cells were significantly increased in iris ciliary-body tissues from LPS injected rats (A, B). In PEA-treated animals, the grade of inflammation (A) and the number of infiltrating cells (B) were significantly reduced. In addition, (C) showed a reduction of MPO activity on iris–ciliary body samples from PEA-treated rats. This effect was comparable with positive control (DEX) (A,B,C). Data are the mean ± S.D. ($n = 10–12$ eyes). *$P < 0.01$ vs. Sham, °$P < 0.01$ versus vehicle.

experiments are representative of at least three experiments (5 slides for each eye from different animals) performed on different experimental days.

3. Results

3.1. Histopathology of EIU and clinical score

Histological analysis of iris ciliary-body tissues from LPS injected animals showed signs of severe uveitis with a strong neutrophil infiltration (Fig. 1B and B1). In epithelial cells from LPS injected rats was observed reduced cytoplasm and visible nucleus (Fig. 1B and B1). In PEA-treated animals, the degree of inflammation was significantly reduced (Fig. 1C and C1) with similar characteristics to the sham-treated rats (Fig. 1A and A1).

The number of neutrophils was also significantly decreased in the group treated with PEA (Fig. 2B) as well as the MPO activity on iris–ciliary body samples (Fig. 2C). In addition, 16 h after LPS injection, the clinical inflammatory score, was significantly reduced in PEA group compared with controls (Fig. 2A). The effect of PEA was comparable with that of the positive control group receiving DEX (Fig. 1D and D1; Fig. 2A–C).

3.2. MDA, protein and TNF-α levels

16 h after LPS injection, increased protein levels were observed in the aqueous humor from LPS injected rats (Fig. 3A). PEA treatment significantly decreased the leakage of proteins in the eye tissue (Fig. 3A). As shown in Fig. 3B the levels of TNF-α in the aqueous humor were significantly reduced in PEA-treated rats compared to the vehicle group. In addition, the levels of lipid peroxidation, measured by MDA levels in ocular tissues, were decreased in PEA-treated rats compared to vehicle group (Fig. 3C). The effect was comparable with that observed with DEX (Fig. 3A–C).

3.3. Effects of PEA on adhesion molecule expression

The results obtained from histological analysis are correlated with immunohistochemical staining for adhesion molecule ICAM-1. The sections from LPS injected rats showed intense positive staining for ICAM-1 (Fig. 4 B and see E). On the contrary, reduced positive ICAM-1 staining was observed in the eyes samples of PEA-treated rats (Fig. 4C and E). A similar effect was also found in the eyes sample treated with DEX (Fig. 4D and E). No staining was observed in sham animals (Fig. 4A and E).

3.4. Effects of PEA on nitrotyrosine and iNOS expression

No positive staining was observed for nitrotyrosine in the iris–ciliary body from sham-treated rats (Fig. 5A and E). Immunohistochemical analysis of tissue sections obtained from LPS injected rats showed positive staining for nitrotyrosine (Fig. 5B and E). Moreover, a reduced positive nitrotyrosine staining was found in the eye sections of mice treated with PEA (Fig. 5C and E) or positive control DEX (Fig. 5D and E) compared to vehicle group In addition, we also evaluated by western blot the expression of iNOS. A significant increase of iNOS expression 16 h after LPS injection was also detected in iris–ciliary body tissues obtained from mice subjected to LPS-induced uveitis (Fig. 6C). PEA treatment significantly attenuated iNOS expression (Fig. 6C).

3.5. Effect of PEA on IκB-α degradation and NF-κB p65 translocation

We evaluated IκB-α degradation and nuclear NF-κB p65 translocation by western blot analysis to investigate the cellular mechanisms

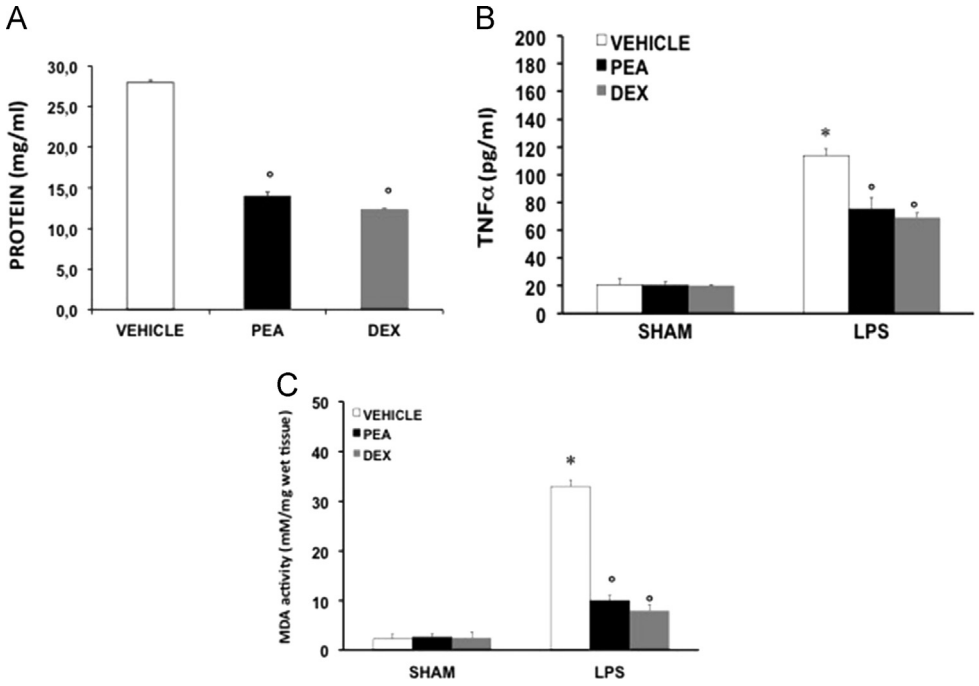


Fig. 3. Effect of PEA on MDA, TNF-α and protein levels. Increased TNF-α (B) and proteins (A) levels were measured in aqueous humor of LPS injected rats. PEA treatment reduced these inflammatory TNF-α (B) and proteins (A) levels. In addition, increased levels of MDA (C) were found in ocular tissues from LPS injected rats. PEA treatment reduced MDA levels (C). This effect was comparable with positive control (DEX) (A–C). Data are the mean ± S.D. ($n=10$–12 eyes). *$P<0.01$ vs. Sham, °$P<0.01$ versus vehicle.

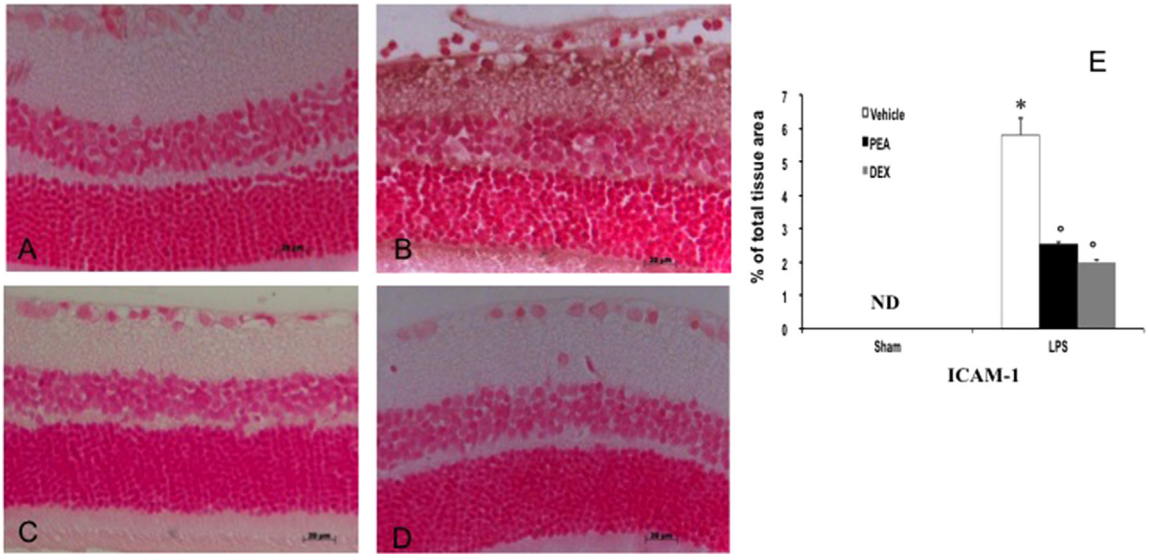


Fig. 4. Effect of PEA on ICAM-1 expression. No positive staining for ICAM-1 was observed in the eyes tissues from sham-treated mice (A and E). On the contrary immunohistochemical analysis for ICAM-1 (B and E) showed positive staining (brown color) in the eyes tissues from LPS-injected animals. The intensity of the positive staining for ICAM-1 was markedly reduced in tissue sections obtained from LPS-injected animals which have been treated with PEA (C, E) or DEX (D, E). Densitometry analysis of immunocytochemistry photographs ($n=5$) for ICAM-1 from eye tissues was assessed (E). The assay was carried out by using Optilab Graftek software on a Macintosh personal computer (CPU G3-266). Data are expressed as percentage of total tissue area. This figure is representative of at least 3 experiments performed on different experimental days. Data are the mean ± S.D. ($n=10$–12 eyes). *$P<0.01$ vs. Sham, °$P<0.01$ versus vehicle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

associated with PEA treatment in LPS- induced uveitis. NF-κB p65 levels in the nuclear fractions of eye tissues were increased at 16 h after LPS injection compared to the sham group (Fig. 6B). Treatment with PEA reduced the levels of NF-κB p65 (Fig. 6B). Moreover, basal expression of IκB-α was found in eye tissues from sham animals, whereas IκB-α levels were substantially reduced in eye tissues obtained from vehicle-treated animals (Fig. 6A). PEA treatment prevented LPS- induced IκB-α degradation (Fig. 6A).

4. Discussion

EIU is an animal model of ocular inflammation which causes a breakdown of the blood aqueous barrier (Bucolo et al., 2003). It develops within 24 h of endotoxin challenge, with signs of uveitis that range from discrete dilatation of the iris and conjunctival vessels to intense iridal hyperemia with flare in the anterior chamber, along with fibrinous exudate in the pupillary area. These

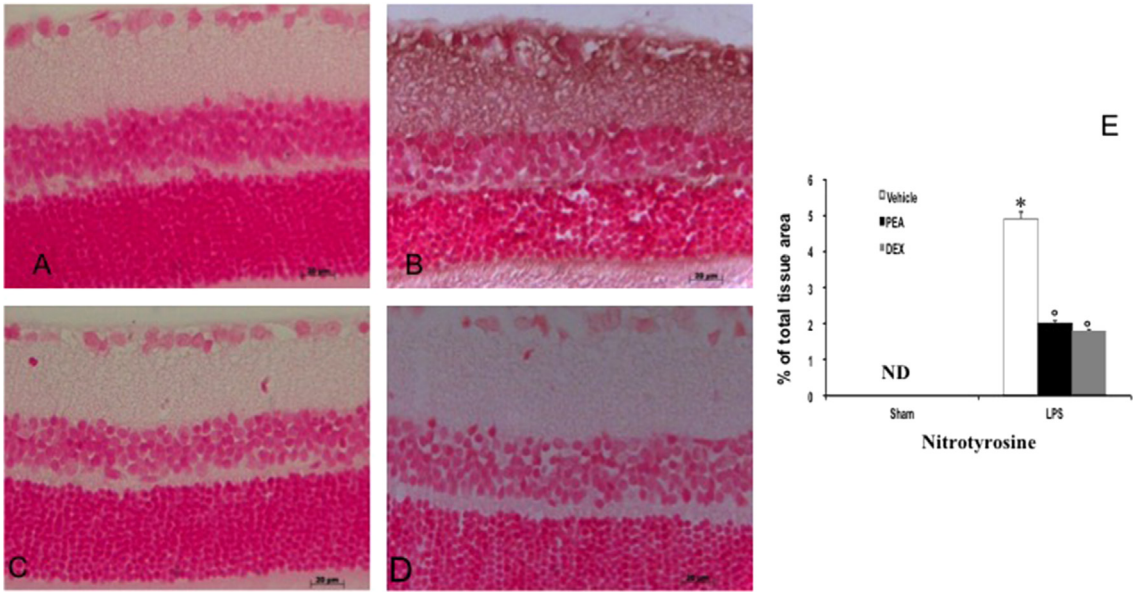


Fig. 5. Effect of PEA on nitrotyrosine expression. No positive staining for nitrotyrosine was observed in the eyes tissues from sham-treated mice (A and E). On the contrary immunohistochemical analysis for nitrotyrosine (B and E) showed positive staining (brown color) in the eyes tissues from LPS-injected animals. The intensity of the positive staining for nitrotyrosine was markedly reduced in tissue sections obtained from LPS-injected animals treated with PEA (C and E) or positive control DEX (D, E). Densitometry analysis of immunocytochemistry photographs ($n = 5$) for nitrotyrosine from eye tissues was assessed (E). The assay was carried out by using Optilab Graftek software on a Macintosh personal computer (CPU G3-266). Data are expressed as percentage of total tissue area. This figure is representative of at least 3 experiments performed on different experimental days. Data are the mean ± S.D. ($n = 10$–12 eyes). *$P < 0.01$ vs. sham °$P < 0.01$ versus vehicle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

clinical signs are accompanied by augmented content of proteins and inflammatory mediators such as interleukin-6, TNF-α and NO in the aqueous humor (Bucolo et al., 2003; Jin et al., 2006; Suzuki et al., 2006). A spontaneous regression of clinical, histological and seroimmunological signs of EIU can be observed starting 16–24 h after LPS challenge, with marked amelioration being observable within 72 h after induction of the disease (Medeiros et al., 2008). In that regard, the aim of this study was to investigate whether PEA, a member of the fatty acid ethanolamine family, could modulate the inflammatory events leading to EIU after injection of LPS. In particular, we propose the following cycle: LPS induced eye inflammation – reactive oxygen species production – NF-κB pathway modulation – proinflammatory cytokines and NO release – endothelial injury – neutrophil infiltration – eye tissue damage. It has previously been shown that specific blockade of NF-κB, TNF-α and NO exerts beneficial effects on the course of EIU (Kitamei et al., 2006; Suzuki et al., 2006). Thus, because of its important pharmacological properties, PEA treatment, by the interaction with this timeline of events could reduce the eye inflammation induced by LPS injection in a rat model of uveitis.

Several studies have demonstrated that PEA has important analgesic, anti-inflammatory and neuroprotective activities (Ahmad et al., 2012; Esposito et al., 2012). A recent study demonstrated that ultramicronized PEA attenuated the renal dysfunction and injury associated with ischemia reperfusion (IR) (Di Paola et al., 2012a). Another study in a compression model of spinal cord injury (Esposito et al., 2011) showed that ultramicronized PEA treatment decreases mast cell (MC) infiltration and activation, which occurs not only during inflammation, but also in neuropathic hyperalgesia. Previous works (Smith et al., 1998) demonstrated that the neutrophil infiltration has a key role in the pathogenesis of EIU (Smith et al., 1998). During inflammatory conditions, adhesion molecules such as ICAM-1 are expressed on endothelium (Roebuck, 1999; Whitcup et al., 1995) and are activated by different cytokines. In that regard, PEA treatment decreased the levels of TNF-α and ICAM-1 expression in various models of inflammation (Di Paola et al., 2012a, 2012b). In the present study we also observed a significant reduction of TNF-α levels in the aqueous humor and ICAM-1 expression in iris–ciliary body sections obtained from rats injected with LPS and treated with PEA. It is possible that the diminished expression of ICAM-1 in the iris ciliary bodies during development of EIU may be attributed to drug-induced downregulation of TNF-α production. In addition, we showed that PEA decreased the levels of proteins in the aqueous humor and neutrophil infiltration, measured by MPO activity, in eyes samples obtained from EIU-subjected rats treated with PEA. Thus, by inhibiting TNF-α induction and intraocular expression of ICAM-1, PEA may have prevented infiltration of the eyes with inflammatory cells and, hence, local production of pro-inflammatory mediators.

Oxidative stress is also suggested to induce eye inflammation (Hollyfield et al., 2008; Sellak et al., 1994). Oxidative biomarkers are elevated in EIU (Satici et al., 2003) suggesting that inflammation and oxidative stress together contribute to its pathogenesis (Kubota et al., 2009). ROS can activate transcription factors such as NF-κB which plays a central role in inflammation (Kitamei et al., 2006; Wang et al., 2002). Under normal conditions, NF-κB is sequestered in the cytosol bound to the inhibitor protein (I)κB-α. Exposure of cells to LPS activates the phosphorylation and degradation of IκB-α. Once IκB-α dissociates from the complex, NF-κB can translocate into the nucleus and can activate inflammatory genes transcription. Our data showed that PEA reduced the translocation of NF-κB p65 in nucleus and the degradation of IκB-α in inflammed eyes tissues. These results are also in agreement with a previous study in which central administration of PEA reduces hyperalgesia via inhibition of nuclear NF-κB pathway in dorsal root ganglia (D'Agostino et al., 2009). Thus, in our work the reduction of TNF-α levels, adhesion molecule (ICAM-1) and neutrophil infiltration could be also related to the the inhibitory effect of PEA on NF-κB pathway. Moreover, the exact mechanisms by which PEA suppresses NF-κB activation are not known; in several inflammatory experimental models, it was related to activation of PPAR-α receptor (Di Paola et al., 2012a) but some other pathways could be involved.

Among free radicals, NO is one of the proinflammatory mediators of EIU (Goureau et al., 1995) which is responsible to the production of peroxynitrite anion (ONOO–). Peroxynitrite can cause lipid

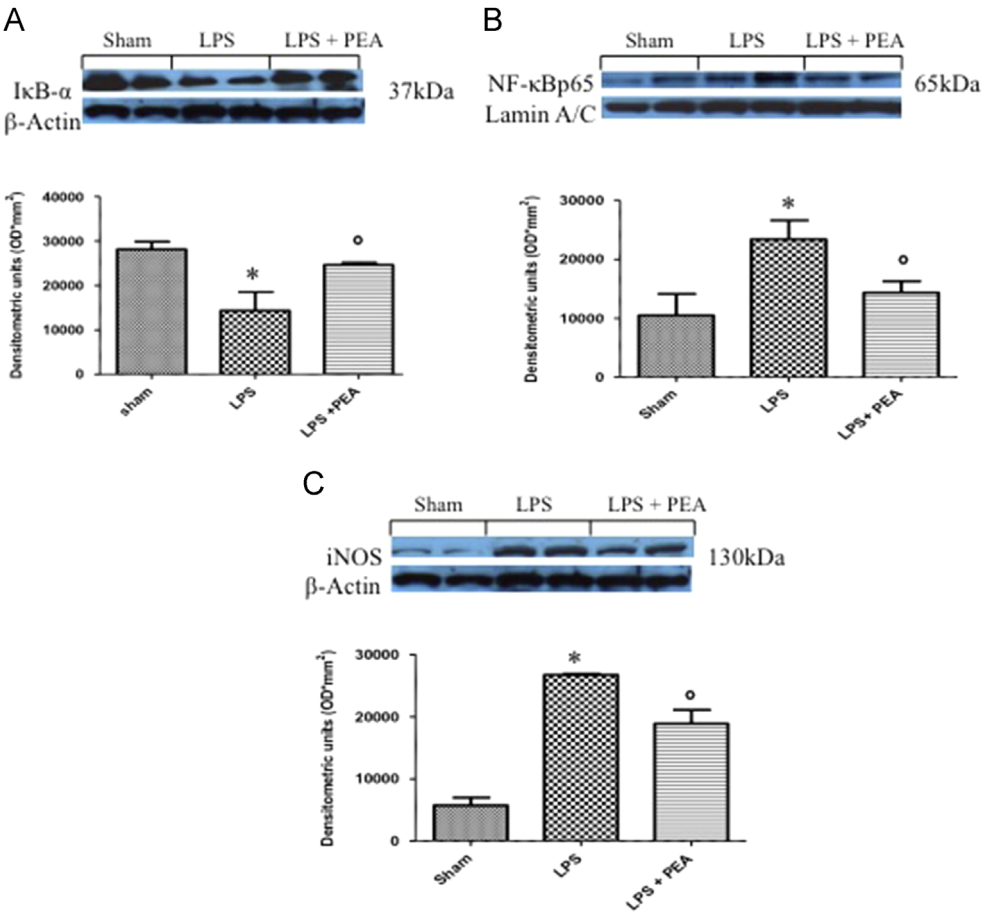


Fig. 6. Effect of PEA on IκB-α degradation, NF-κB p65 translocation and INOS expression. By Western blot analysis NF-κB p65 levels in the eyes nuclear fractions were significantly increased after LPS injection compared to the sham-treated mice (B). PEA treatment significantly reduced the levels of NF-κB p65 (B). Basal expression of IκB-α was detected in eyes samples from sham-treated animals, whereas IκB-α levels were substantially reduced in eyes tissues obtained from vehicle-treated animals after LPS injection (A). PEA treatment prevented LPS-induced IκB-α degradation (A). In addition, an increase of iNOS expression was observed in eyes tissues from LPS injected rats (C). This expression was reduced by PEA treatment (C). The relative expression of the protein bands was standardized for densitometry analysis to b-actin and lamin a/c levels. Data are the mean ± S.D. ($n = 10–12$ eyes). *$P < 0.01$ vs. sham, °$P < 0.01$ versus control vehicle.

peroxidation; infact, in this study we also showed that PEA treatment decreased lipid peroxidation by measurement of MDA levels. Therefore, a recent study also showed that PEA treatment protects the kidney from hypertensive injury by the inhibition of oxidative stress (Mattace Raso et al., 2013). Peroxynitrite is involved in the nitration of tyrosine residues (nitrotyrosine) leading protein structure and function changes. Increased nitrotyrosine staining is considered as an indicator of "increased nitrosative stress". In this study, we demonstrated that EIU leads to a significant increase of nitrotyrosine and iNOS formation. Here, we demonstrated that PEA treatment was able to reduce nitrotyrosine staining and iNOS expression. Moreover, a previous study has shown that PEA administration significantly reduced the degree of spinal cord inflammation and tissue injury, pro-inflammatory cytokine expression, neutrophil infiltration, nitrotyrosine formation, NF-κB activation and apoptosis (Genovese et al., 2008).

In conclusion, PEA is a protective endogenous mediator produced during inflammatory conditions to counteract inflammation, neuronal damage and pain. This study demonstrates that PEA treatment is able to attenuate the eye inflammation induced by LPS in the rat. Importantly, PEA was effective in counteracting the uveitis-inducing effects of LPS, when administered, 1 h before and 7 h after challenge with LPS. The later time is a point when inflammatory responses, including TNF-α and NO release in the aqueous humor, have already commenced (Medeiros et al., 2008). In addition, in our studies, in spite of the well-known efficacy of steroidal therapy in uveitis patients, the same results were obtained for the positive controls DEX-treated rats.

Thus, these protective effects of PEA may be due to the inhibition of some specific cytokines which are under the control of NF-κB pathway. Anyway, the results of the present study suggest that PEA could have an important role as therapeutic target in ocular inflammation and could be an interesting agent for the treatment of human uveitis.

Conflict of interest

The authors declared no conflict of interest.

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